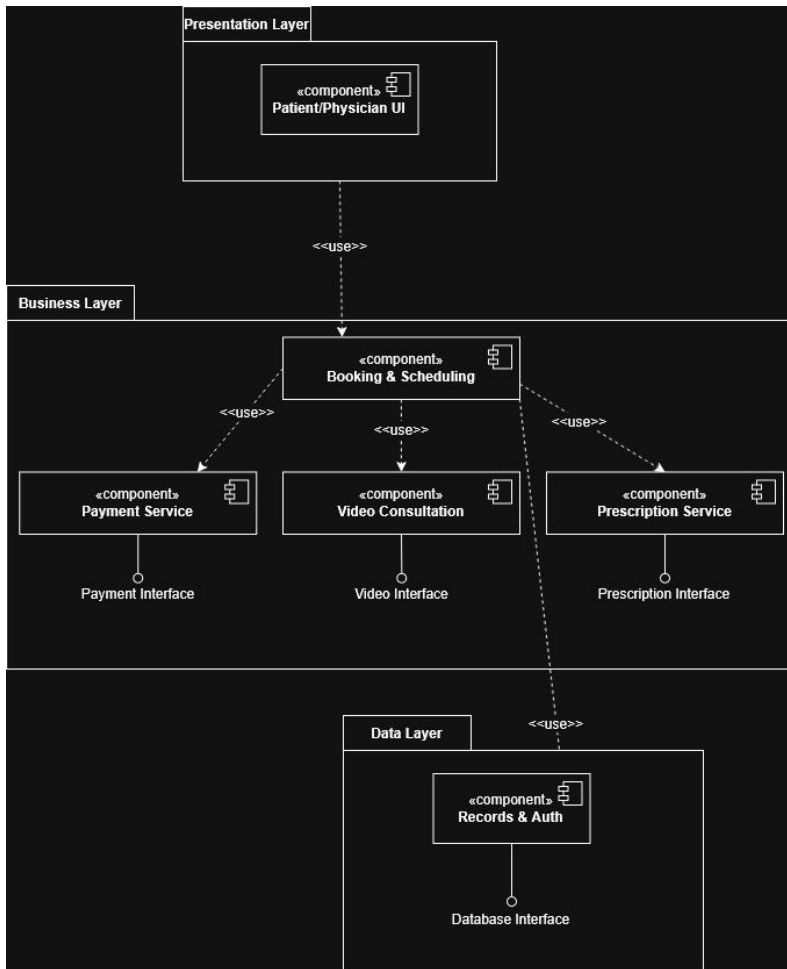


Architecture Justification: Telemedicine Portal

We chose Layered Architecture for the Telemedicine Portal system.



Why we picked this architecture

Reason 1: The system naturally splits into three clear jobs — what the user sees (Presentation layer), what the system decides and does (Business layer: booking, payment, video calls, prescriptions), and where the data is kept (Data layer: patient and doctor records). Layered Architecture matches this split directly, so the design is easy to explain and easy to build.

Reason 2: Each layer only talks to the layer next to it. If we later change how the booking screen looks, we do not need to touch the payment or database code. This keeps the system simple to test and fix, one layer at a time, which fits well within our project timeline.

Security advantage

Because the Data layer is only reachable through the Business layer, patient records and prescriptions are never exposed directly to the Presentation layer. Every request for sensitive health data has to pass through the business rules first, so we can check permissions and encrypt data (as required by NFR-001) in one place instead of many.

Performance benefit

Keeping the Video Consultation component separate from Booking and Payment inside the Business layer means a slow video connection does not block a patient from booking a slot or paying. Each component can be optimized on its own, which helps us meet the 5-second video connection target (NFR-002) without redesigning the whole system.